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I build the app your customers use, the app your staff run the business from, and the backend that keeps both honest.

Four products, built end to end: a queuing and booking platform for barbershops and salons, a running-community platform with live GPS events, an online pharmacy with delivery, and a coffee-shop ordering system (the last two are still in progress). Each one is a full system — a customer app, an operations app, and a database with its own security rules, payments, push notifications and monitoring — built and maintained by me.

4 products, each a full system	**7** Flutter apps for iOS & Android	**670k** lines of hand-written Dart	**1,860+** commits since Aug 2023
246 Postgres migrations (schema, RLS, RPCs)	**37** server functions (Supabase Edge + Firebase)	**300** Flutter test files	**79** database test suites (pgTAP)

Counted from the four repositories on 25 Oct 2026. Generated code (*.g.dart, *.freezed.dart) excluded.

SELECTED WORK

Four products, seven apps

Every project has two audiences: the people who buy, book or run, and the people who serve them. I build for both, on one shared backend.

PRODUCT	WHAT IT DOES	APPS	BACKEND	SINCE
Dorey	Queuing & booking that cuts waiting time at barbershops and salons	Customer · Business (owners, managers, receptionists, admins)	Supabase	2023
EgyRuns	Running community: clubs, events, GPS tracking, live race-day operations	Runner · Mgmt (organisers, staff, admins)	Supabase	2026
Sehety IN PROGRESS	Online pharmacy with prescriptions, wallet and delivery	Customer · Mgmt (admin, doctor, moderator, driver)	Supabase	2026
Chicco IN PROGRESS	Coffee-shop ordering, custom drinks, loyalty and delivery	One app, six roles (client, barista, delivery, manager, admin, partner)	Firebase	2025

WHAT I DO

The whole product, start to finish

From the first screen to the server behind it and the store release, everything is built to work together, with no gaps between the app and the backend.

MOBILE

Customer apps

Flutter apps for iOS and Android with Arabic and English (right-to-left), maps, payments, push, home-screen widgets and iOS Live Activities.

OPERATIONS

Staff & admin apps

Role-based apps for owners, managers, drivers and support: dashboards, catalog editing, scanning, approvals, exports to PDF, CSV and Excel.

BACKEND

Database & APIs

Postgres on Supabase: schema, row-level security, server functions for anything involving money or stock, Edge Functions, scheduled jobs.

TRANSACTIONS

Orders & payments

Flows that survive bad networks and double taps: holds, idempotent writes, server-side price checks, wallets, loyalty points, promo codes.

RELIABILITY

Production readiness

CI pipelines, force-update gates, feature flags, crash and error logging, uptime checks, service-level objectives and incident runbooks.

MIGRATION

Rescue & re-platform

Moving live apps between stacks: Firestore to Postgres, Provider to Riverpod, single repo to a shared-package monorepo, without breaking old app versions.

HOW A PROJECT RUNS

From first call to running in production

The order below is how every project above was built. You see working software early and every week after.

Step 1

Discover

Who uses it, who runs it, and where money or time is lost today. Written up as screens and roles.

Step 2

Model

Tables, permissions and the rules that must never break (no double booking, no negative stock).

Step 3

Build

Customer and staff apps grow together on one backend, with a test build to try every week.

Step 4

Harden

Retries, offline cases, security review, load on the busy paths, and automated tests.

Step 5

Launch & run

Store release, monitoring, force-update and feature flags, then ongoing support.

Personal care · Booking
Aug 2023 → present
937 commits

Dorey

A queuing app that gets people into the chair on time instead of into a waiting room.

Customers find a barbershop or salon, pick a service and a barber, and take a real slot that nobody else can take. Shops run their whole day from a separate business app: branches, staff schedules, the check-in desk, promotions and reports. Built generic enough to take on other service verticals later.

CUSTOMER APP

Browse, book, show up

- Quick search across shops, branches and services (fuzzy, trigram-indexed)
- Slot picker that updates live as other people book, for one barber or "any barber"
- Hold → confirm booking flow: the slot is held for 5 minutes while the customer finishes, with deposits wired in
- Phone sign-in with OTP, bookmarks, booking history, reviews
- Reminder an hour before, cancellation, and notices when the shop changes a booking
- Arabic and English, including notification text

BUSINESS APP

Owners, branch managers, receptionists, admins

- Branches, barbers, services, prices and weekly schedules with breaks
- Check-in desk and no-show handling, with a 24-hour undo
- Editing a schedule previews which bookings it would cancel before saving
- Promo codes per branch, shop clients, staff accounts with approval flow
- Dashboards and deep analytics: bookings, revenue and finance over any date range
- Bulk import of barbers and services from spreadsheets

BACKEND

Supabase · 84 migrations · 5 Edge Functions

- Postgres with row-level security on every table; roles minted into the login token by a database hook
- Booking server functions: `hold_booking`, `confirm_booking`, `create_booking`
- Push notifications through a queue with retries and a dead-letter list
- Edge Functions for OTP, push, geocoding, storage clean-up
- 37 database test suites, run in CI

PROBLEMS SOLVED

No double bookings

Two customers can never get the same barber at the same time, even if they tap at the very same second. The second person is told straight away that the time is taken.

No duplicate bookings or charges

If a customer's connection drops while booking and they try again, they get the booking they already made, never a second one.

Staff can't overwrite each other

If two managers edit the same shop at once, the second is warned instead of silently erasing the first person's changes.

Safe schedule changes

Before a shop changes its hours, the app shows exactly which bookings would be affected, and those customers are notified.

Availability that is always current

Open times update on screen as other people book, so customers don't choose a time that has already gone. It stays fast and affordable even when a busy shop has many people looking.

Fixes without store delays

A new feature can go to a few customers first, or be switched off within minutes, with no app store review. Errors are reported automatically, so issues are fixed before customers complain.

`Flutter` `Riverpod` `GoRouter` `Supabase` `Postgres RLS` `pgmq` `pg_cron` `Edge Functions (Deno)` `FCM` `Melos monorepo` `slang i18n` `GitHub Actions`

Sport · Community · Events

2026 → present

2 apps + shared packages

EgyRuns

Egypt's running scene in one app, and the race-day console that runs the events.

Runners discover clubs and events on a map, register, track runs with GPS and see their results. Club organisers and event staff use a second app to create runs, plan courses, check people in at the gate, manage sponsors and booths, and watch the event live.

RUNNER APP
Discover, register, run

- GPS run tracking that keeps going in the background, with an audio coach
- iOS Live Activities and home-screen widgets during a run
- Bluetooth heart-rate sensors, Apple Health / Health Connect import
- Garmin and WHOOP account sync
- Event map, clubs, friends, leaderboards, run analysis charts
- 3D course fly-through of the race route; route builder snapped to real paths
- Event registration forms, QR ticket, results and podium

MGMT APP
Club owners, event staff, admins

- Clubs, rosters, member bans, verified-club badge
- Run creation with duplicate, route library, course records
- Marshal / aid-station / photographer roles and aid-station markers
- Live event console with a map of runners
- Bib and gate QR scanning, sponsor booths and redemptions, kit pick-up
- Money view, staff accounts, push notifications, PDF and CSV exports

BACKEND
Supabase · 136 migrations · 13 Edge Functions

- RPCs for every former transaction; counters owned by triggers, not clients
- OAuth start, callback, webhook and token refresh for wearables
- Vendor tokens encrypted at rest (AES-GCM)
- Heatmap export, analytics snapshots, ID-image clean-up
- 42 database test suites

ALSO BUILT

A 30-second vertical ad film for the runner app, made in code with Remotion and rendered to 1080×1920 or 4K from the same app screens.

PROBLEMS SOLVED

Runs are never lost

Tracking keeps going with the screen off or the phone in a pocket, and the run is saved as it happens, so a crash doesn't erase it.

Works with runners' watches

Runners link their Garmin or WHOOP once and their workouts appear on their own. Account details are kept securely on the server, never on the phone.

Quick check-in on race day

Staff scan bibs and tickets at the gate and at sponsor booths and get a clear yes or no on screen, built for crowds and weak mobile signal.

See the race route before race day

Runners can fly through the course in 3D, with the elevation, so they know what's coming.

Organisers see the race live

During an event, organisers follow runners on a live map from their phone.

Course records kept automatically

When a runner sets the best time on a club route, the record updates by itself, with nothing entered by hand.

Flutter · Riverpod · Supabase · Mapbox · OSRM · BLE · HealthKit / Health Connect · Live Activities · Garmin · WHOOP OAuth · pgTAP · Remotion · GitHub Actions

IN PROGRESS

Health · E-commerce · Delivery

Feb 2026 → in development

Customer + 4 staff roles

Sehety

An online pharmacy that remembers your medication and brings it to your door.

Customers search or scan a medicine, upload a prescription, keep a list of their regular medication and a vault of medical documents, and pay by wallet or cash. Admins, doctors, moderators and drivers each sign in to one management app and see only their own tools.

CUSTOMER APP

Order, track, refill

- Search with popular searches, category grid, filters by price, brand and symptom
- Barcode scanning, request an unavailable medicine, upload a prescription
- "My medications" list for one-tap refills; product bundles
- Cart suggestions from reorder history and frequently-bought-together
- Checkout with live stock check, wallet or cash, map pin for the address
- Live order tracking, cancellation, loyalty points convertible to wallet
- Medical document vault with folders, PDF and image previews

MGMT APP

Admin · doctor · moderator · driver

- "Needs attention" dashboard, order pipeline with sound alerts for new orders
- Catalog: medications, categories, manufacturers, suppliers, units, Excel import
- Home-screen builder, ads, curated popular searches, bundles
- Wallets with audited balance adjustments, promo codes, best sellers, reports
- Driver shifts and a driver order view; doctor and moderator dashboards
- Role management with a gate that sends each person to their own area

BACKEND

Supabase · 26 migrations · push function

- Money, stock and role changes only through server functions, never direct writes
- Row-level security per role; schema changes linted and checked by advisors
- Push notifications through an Edge Function bridge to FCM

PROBLEMS SOLVED

No surprises at checkout

Stock is checked just before an order is placed. If something has just sold out, the cart updates and the customer sees why, instead of the pharmacy calling them later.

Every pound accounted for

Every wallet change, points conversion and manual adjustment is recorded, so the customer and the admin can always see where a balance came from.

One app for the whole team

Admins, doctors, moderators and drivers use one app, and each person sees only their own tools. Access is also enforced on the server, not just hidden in the app.

Refills in one tap

Customers keep a list of their regular medication and reorder it without searching again.

Flutter · Riverpod · Supabase · Google Maps · mobile_scanner · Live Activities · Home widgets · easy_localization · Excel import · FCM · Melos monorepo

IN PROGRESS

Food & drink · Ordering

Jul 2025 → in development

658 commits

Chicco

A coffee shop's whole operation, from the custom latte to the delivery door.

Customers build their own drinks from real ingredients, pay by card, Apple Pay or wallet, and collect points. Baristas work an order queue, drivers deliver and confirm with a code, and managers see revenue, stock and expenses, all in one app that changes with the signed-in role.

CLIENT
Order and earn

- Menu by category and size; custom blends from ingredients and toppings
- Sign in with Google, Apple, Facebook or phone OTP
- Card payments through Paymob, Apple Pay, wallet top-up
- Points and rewards, favourites, reviews, QR codes
- Live order tracking and delivery address on the map

STAFF ROLES
Barista · delivery · manager · admin · partner

- Barista order queue with preparation details
- Delivery assignments, navigation and code-verified hand-off
- Products, ingredients, topping stock, sizes, stores, cities and delivery fees
- Promo codes, segmented push campaigns, reward programmes
- Revenue analytics, recurring expenses and a daily analytics digest

BACKEND
Firebase · 18 Cloud Functions

- Paymob webhook and wallet payment functions
- Points awarded only after payment is confirmed server-side
- Delivery OTP verification, role claims, promo redemption
- Firestore security rules, App Check

PROBLEMS SOLVED

Payments that can't be faked

An order is only marked paid once the payment company confirms it, so nobody can trick the app into a free order. Loyalty points are only given after payment.

Orders reach the right person

The driver can only complete a delivery with a code from the customer.

No ordering what's sold out

When an ingredient runs out, the drinks that need it show as unavailable before they reach the cart.

Six roles, one app

Customers, baristas, drivers, managers, admins and partners all use the same app. Each sees only what their role allows, and the server checks it too.

Flutter · Riverpod · Firebase · Cloud Functions · Firestore rules · Paymob · Apple Pay · Google Maps · fl_chart · QR

WHAT YOU GET BY DEFAULT

Standards that come with every build

Each of these is already running in the projects above. They come with every new build, not as extras.

Security in the database	Row-level security on every table; the server re-checks the role on every request.	**Separate environments**	Dev, staging and production projects and app flavours, so testing never touches real customers.
Tested where it matters	Unit and widget tests for app logic, database tests for policies and server functions, all in CI.	**Clean, readable code**	Layered architecture, design tokens instead of hard-coded styles, file-size limits, zero analyzer warnings.
Arabic and English	Right-to-left layouts and every string translated, including push notifications.	**Updates you control**	Minimum-version gate and server-side feature flags to switch features off without a store release.
Errors you can see	Crash and error reports per app version, failed background jobs and push deliveries surfaced to admins.	**Documented hand-over**	Architecture notes, release checklists and runbooks live in the repository you own.

TOOLS

Stack

MOBILE

Flutter & Dart

Riverpod, GoRouter

Freezed, json_serializable

Melos monorepos

iOS widgets & Live Activities

BACKEND

Supabase (Postgres, Auth, Storage, Realtime)

Row-level security, SQL functions

Edge Functions (Deno / TypeScript)

Firebase (Firestore, Functions, FCM)

pgmq queues, pg_cron

INTEGRATIONS

Paymob, Apple Pay

Google Maps, Mapbox, OSRM

Garmin, WHOOP, HealthKit

Bluetooth LE sensors

Google, Apple, Facebook sign-in

DELIVERY

GitHub Actions CI/CD

pgTAP, flutter_test

TestFlight & Play Console releases

Remotion (video in code)

Flavours for dev, staging, prod

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